19. SUMMARY OF VASCULAR STRUCTURING BY THE AMNIOTIC CAVITY

DURATION : 05:54 STUDY SHEET

COURSE SUMMARY

This video deeply explores vascular structuring by the amniotic cavity, highlighting its integration with the vitelline system and its role in embryonic development. Key concepts discussed include the movement of resorption towards the centre of the embryo, arterial integration, and the importance of the amniotic cavity in redefining internal spaces. Essential movements such as cephalisation, embryonic coiling, and fluid repatriation are discussed, while emphasising the impact of the notochordal system and the neural tube. This knowledge is crucial for practitioners seeking to recognise the internal dynamics that influence embryonic health and overall well-being.

SUMMARY OF VASCULAR STRUCTURING BY THE AMNIOTIC CAVITY

THE ORIGIN AND INTEGRATION OF THE VASCULAR SYSTEM

The **vascular system**, at its periphery, is strongly integrated into the **vitelline system**. The **amniotic cavity**, initially the same size as the vitelline vesicle, plays a fundamental role.

The embryo elongates and grows considerably. The importance of this amniotic cavity lies in its ability to completely integrate the vitelline system. This movement is crucial and constitutes the central point of our study.

THE RESORPTION MOVEMENT AND ZONE B

In this movement, the entire vitelline system resorbs towards the centre of the embryo, reintegrating the whole. This amniotic cavity, which can be referred to as **Zone B**, in relation to the aortic system, draws everything towards the centre.

This process occurs on both a **transverse abdominal** and **thoracic** plane.

With the growth and establishment of the heart, as well as the significant development of the brain, the amniotic cavity takes over. It completely encompasses the entire space of the **external coelom**. Only a part of this internal coelom will be integrated into an internal coelom. The more the amniotic cavity grows, the more the system integrates.

ARTERIAL INTEGRATION AND EMBRYONIC COILING

Two key elements summarise the integration of the arterial system, from the periphery to the centre: **cephalisation** and **embryonic coiling**. This includes cardialisation, diaphragmatisation, and hepatisation.

The amniotic cavity acts as an integrating movement, redefining spaces. In this dynamic, both horizontally and vertically, Zone B is constructed. The vitelline vesicle, containing primitive information at the venous level, interacts with the exterior to reintegrate the body and the venous system.

THE ROLE OF FLUIDS AND THE VITELLINE VESICLE

The secret lies in the **repatriation of fluids** towards the centre, with the fascial system playing a predominant role in this release.

The vitelline vesicle is a great source of information. The integration movement includes:

- The transformation of the external coelom into an internal coelom.
- The closure of the neural tube.
- Integration and movement towards the centre.
- Construction, orchestrated by the cavity.

RESTORING THE CAVITY AND ZONE B

The objective is to restore this cavity and Zone B to reintegrate the internal circulation.

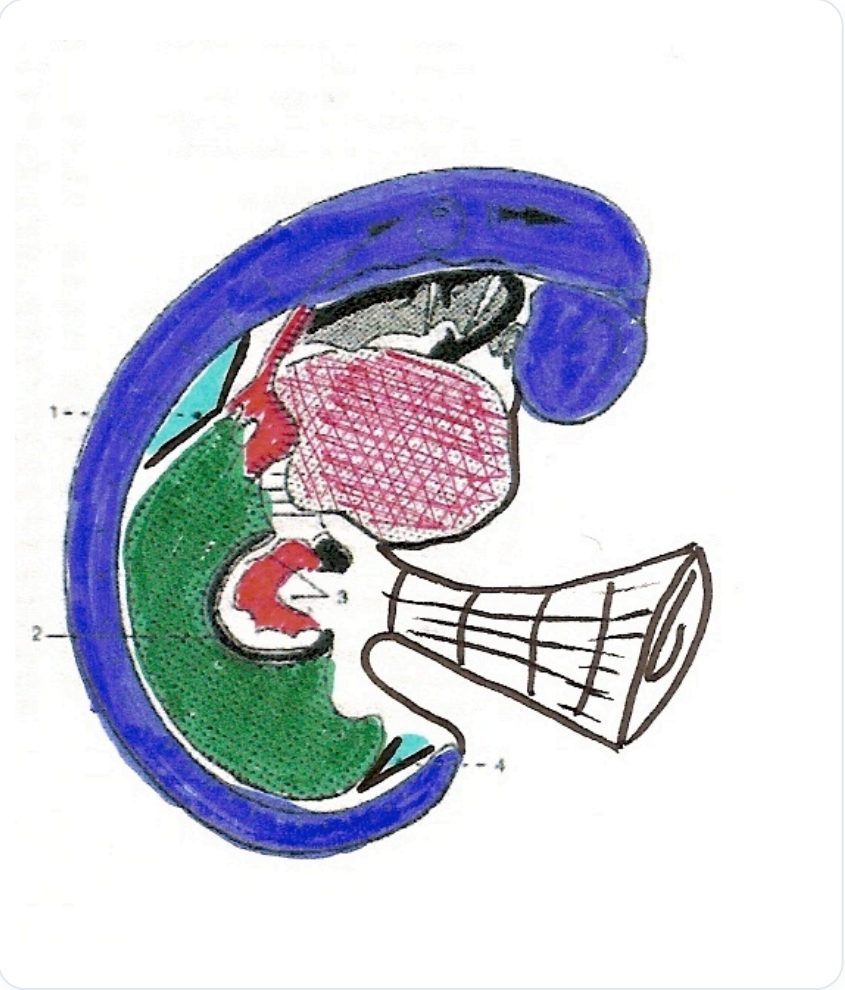


FIG. — This image appears to be an overview, so inserting it after a general contextualizing sentence about the reintegration of the circulatory system seems appropriate

Visualising the diagram as a person lying down:

- In blue, integration into a neural tube.
- The rest forming the skin, an **ectodermal** layer.

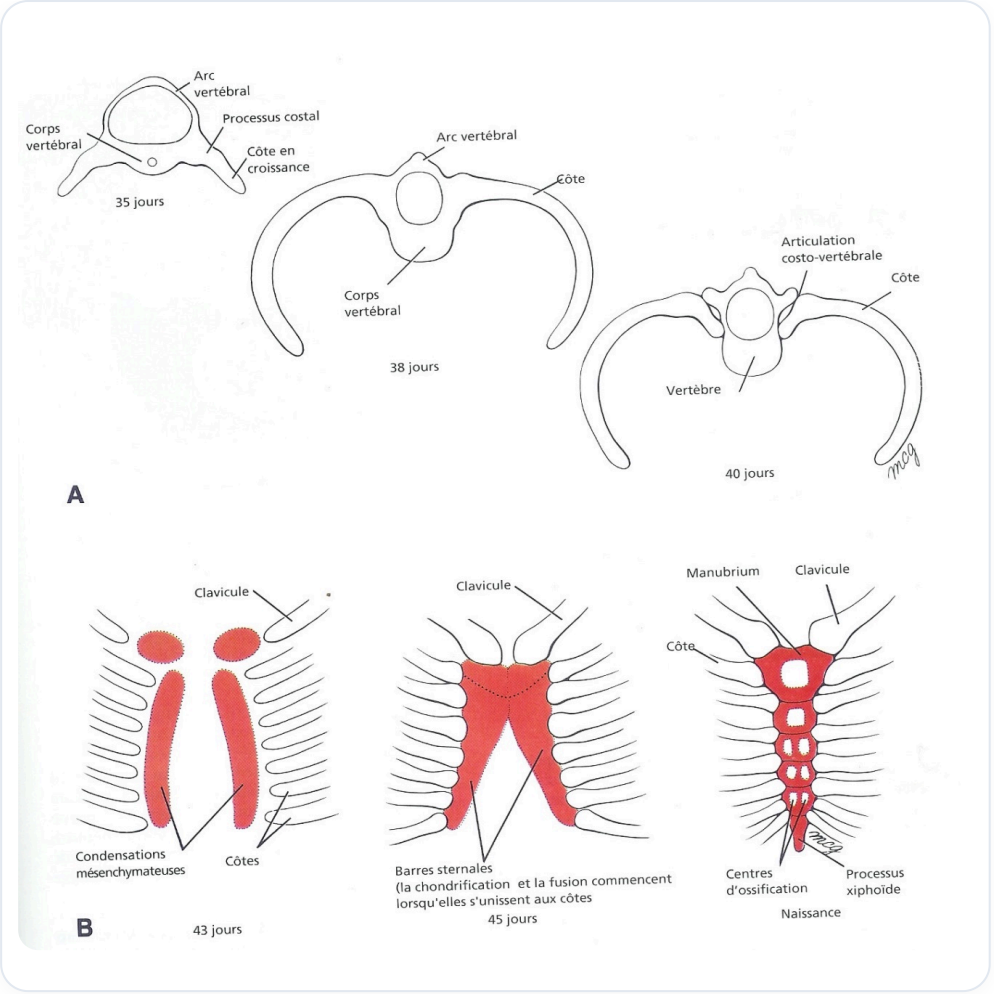


FIG. — The image seems to detail the integration of the nervous system or an aspect of embryonic structuring

Simultaneously, the integration of the vitelline vesicle contributes to forming part of the digestive system, while the entire venous system integrates. The movement to follow is that of the growing amniotic cavity, leading to a fusion to form the longitudinal part, the digestive tube.

The integration of the external coelom into a future internal coelom is visualised in yellow, where everything converges on the midline.

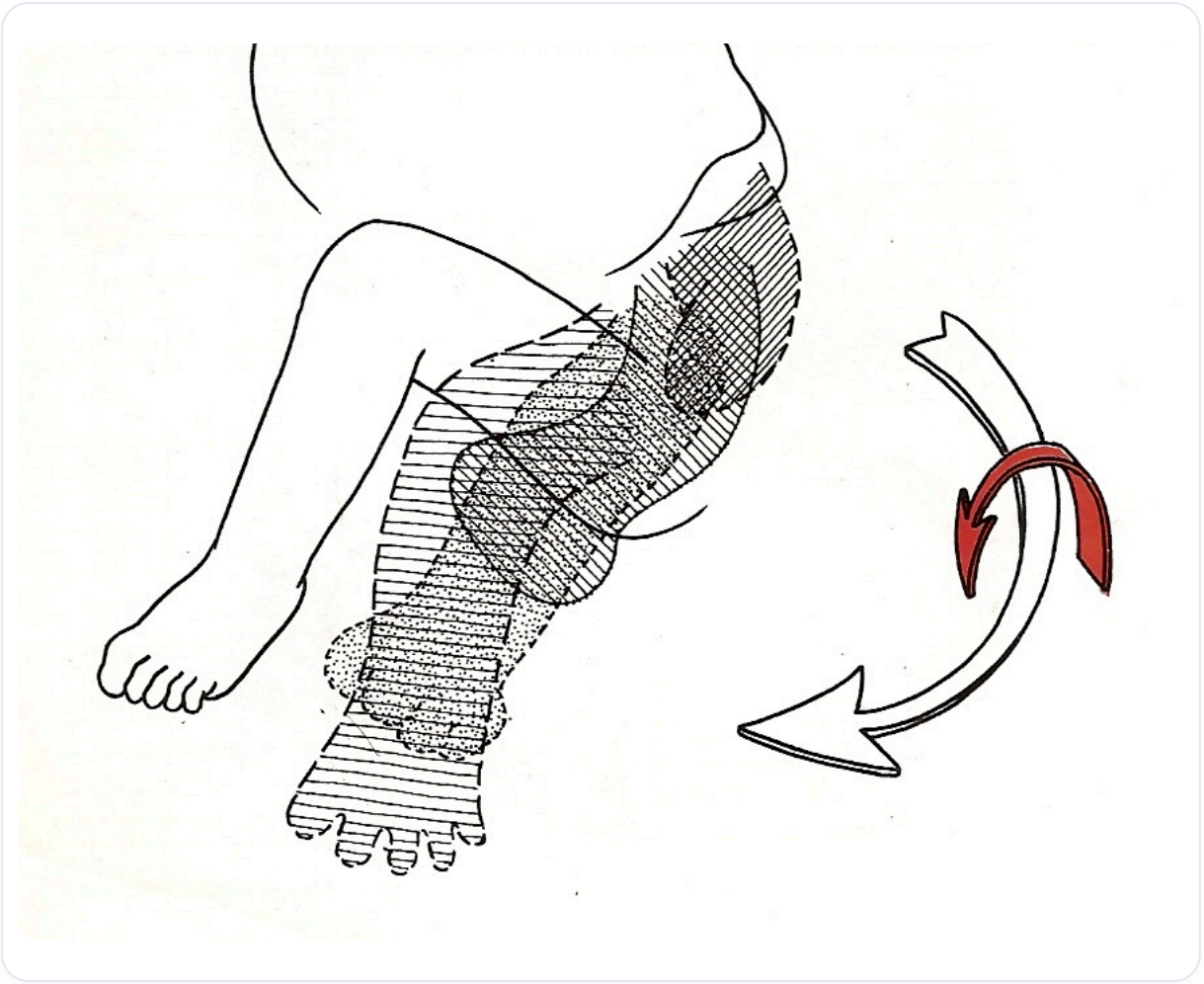


FIG. — Probably a stage of digestive tract formation or a longitudinal structure

THE GREAT ENGINE: NOTOCHORDAL SYSTEM AND NEURAL TUBE

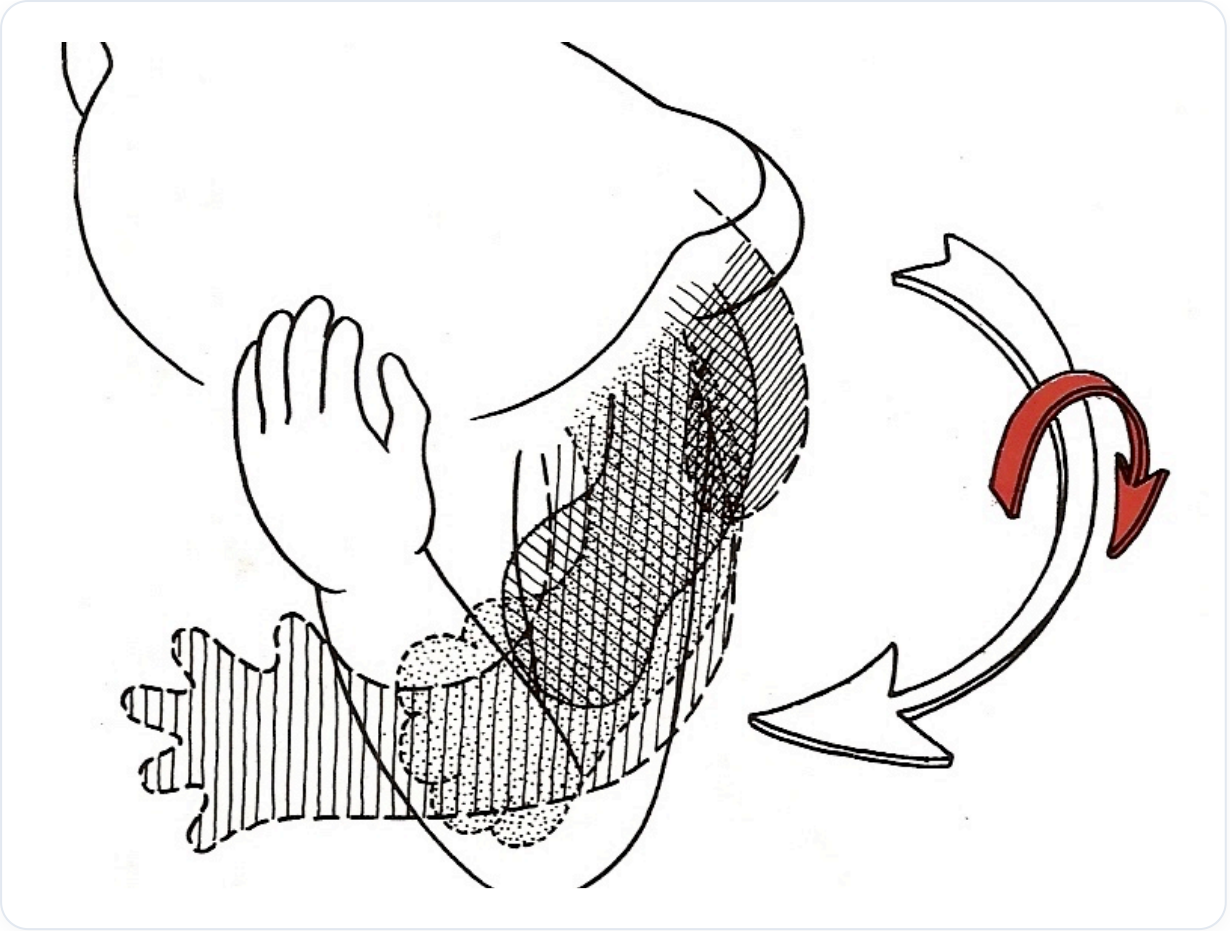


FIG. — Probably the convergence of structures towards the midline, consistent with the sentence

The main engine of this process is the **notochordal system**, which influences the neural tube and organises the primitive aortic vascular system. It redirects information with the progressive integration of the entire transverse system, from the periphery to the centre.

We will follow this movement step by step. It is essential to "allow" the primary respiration to unfold.

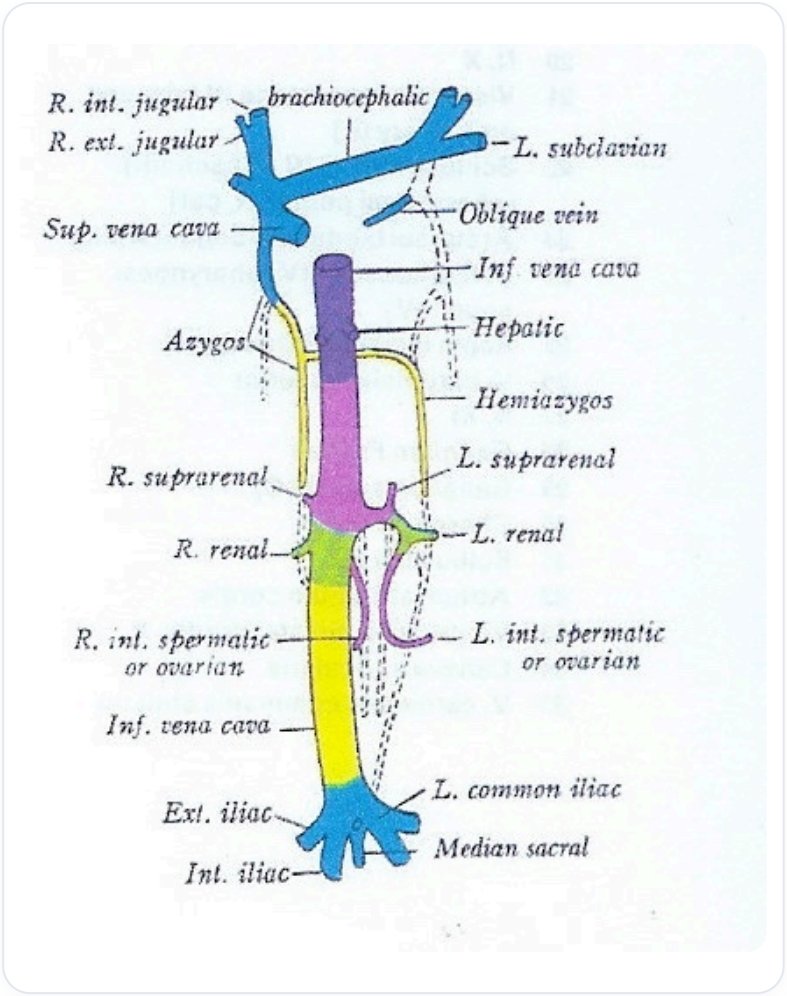


FIG. — Notochordal system or a transverse view of integration, corresponding to the text

ENDOVOLUNTARY MOVEMENT OF THE LUNGS AND AWARENESS OF THE UMBILICAL PEDICLE

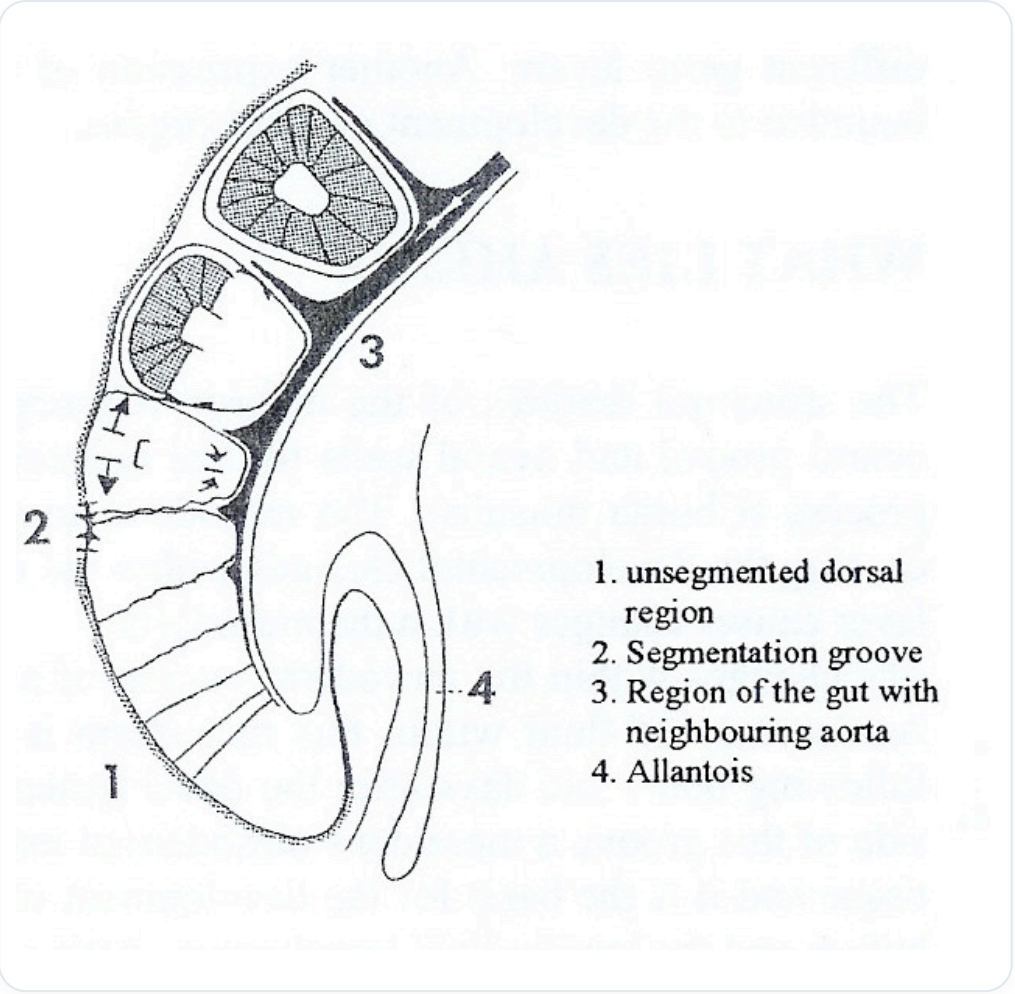


FIG. — The image corresponds to the description of the integration of systems such as the vitelline and hepatic, and this sentence marks a transition to the notion of 'movement'

The **endovoluntary movement of the lungs** originates from the endoderm and is guided by the vascular system. Awareness of the **umbilical pedicle** must return to this space, allowing one to perceive the force of the growing amniotic cavity, while integrating the processes of cephalisation, cardialisation, diaphragmatisation, and hepatisation.

The return then occurs. The vitelline vesicle, propelled by differential growth speed, will be reincorporated into the scheme. It will push the pedicle to join the vitelline vesicle and later form the umbilical cord.

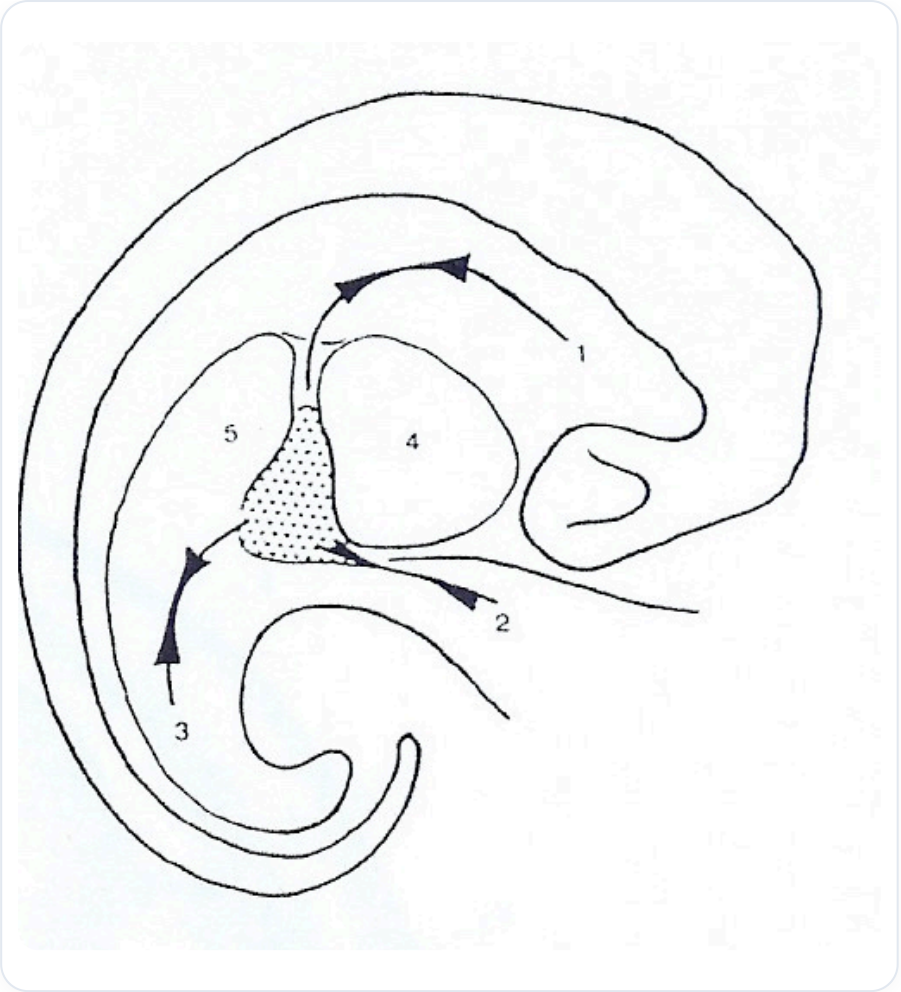


FIG. — The image corresponds to the description of the integration of systems like the vitelline and hepatic

Everything converges here, starting from a pedicle to return to a cord.

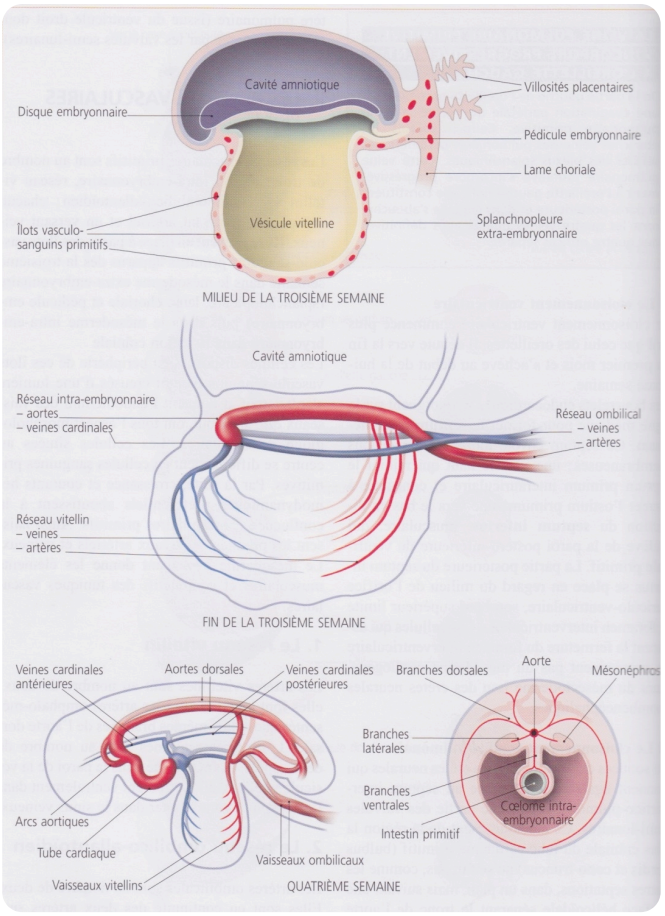


FIG. — Regression of the vitelline vesicle and umbilical cord formation, which is logical after mentioning the 'return'

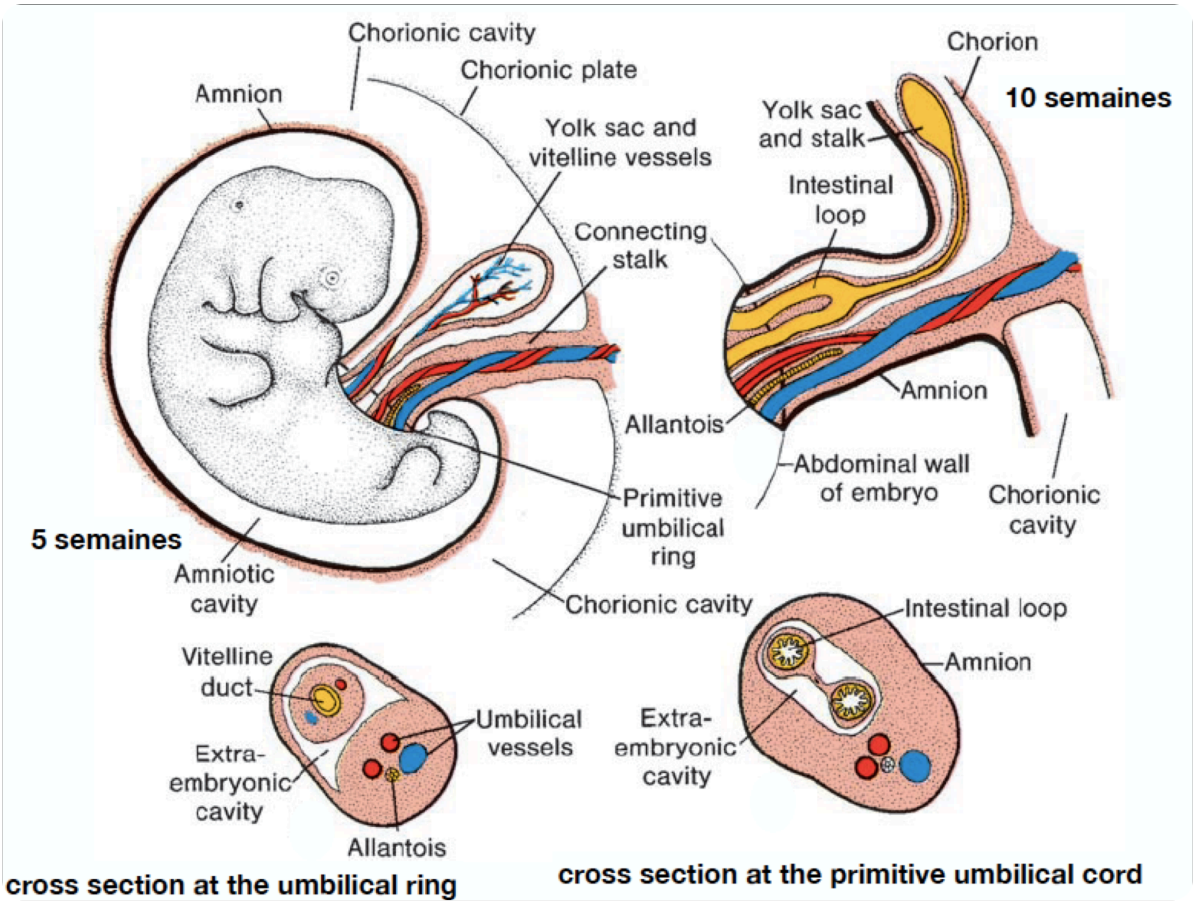


FIG. — Umbilical cord formation

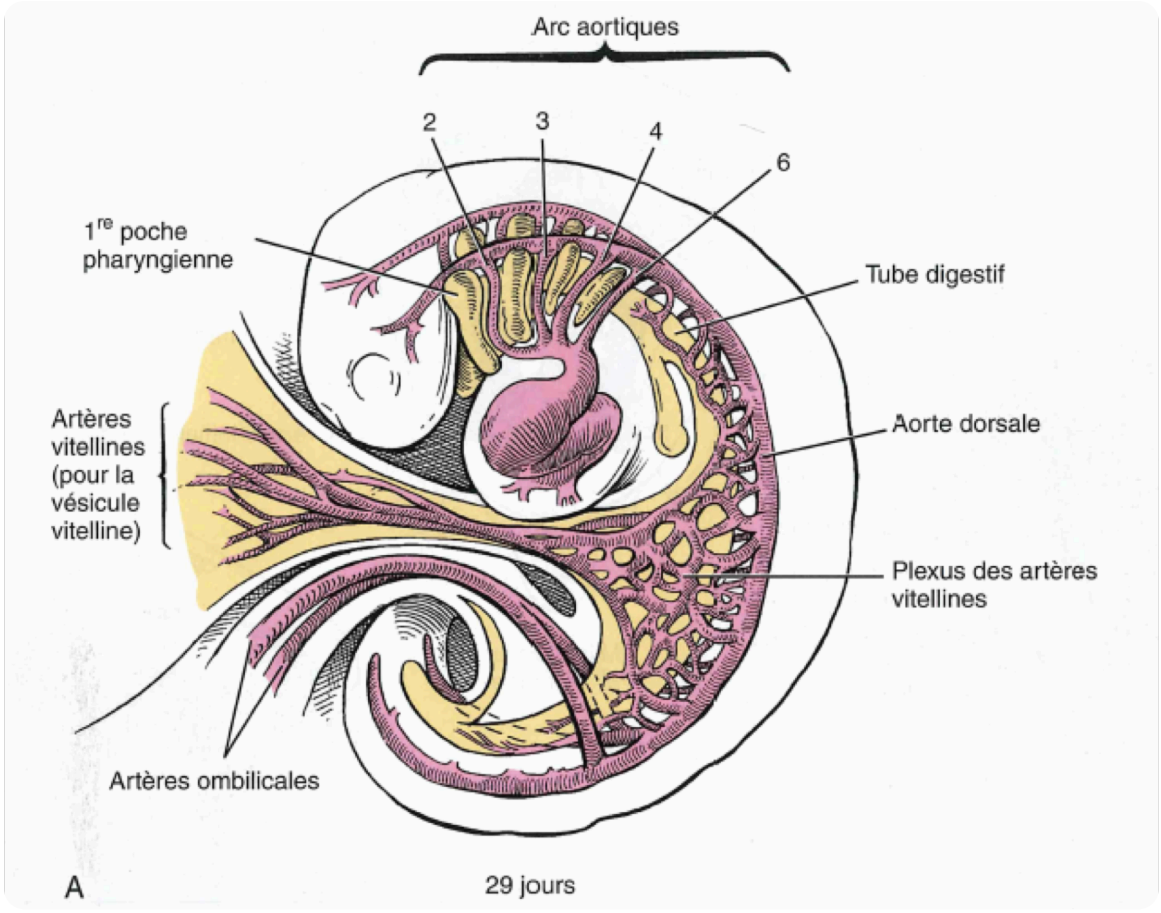


FIG. — This image appears to be a recapitulative view or a final visualization of the integration of the vitelline system and its remnants, consistent with the end of the paragraph

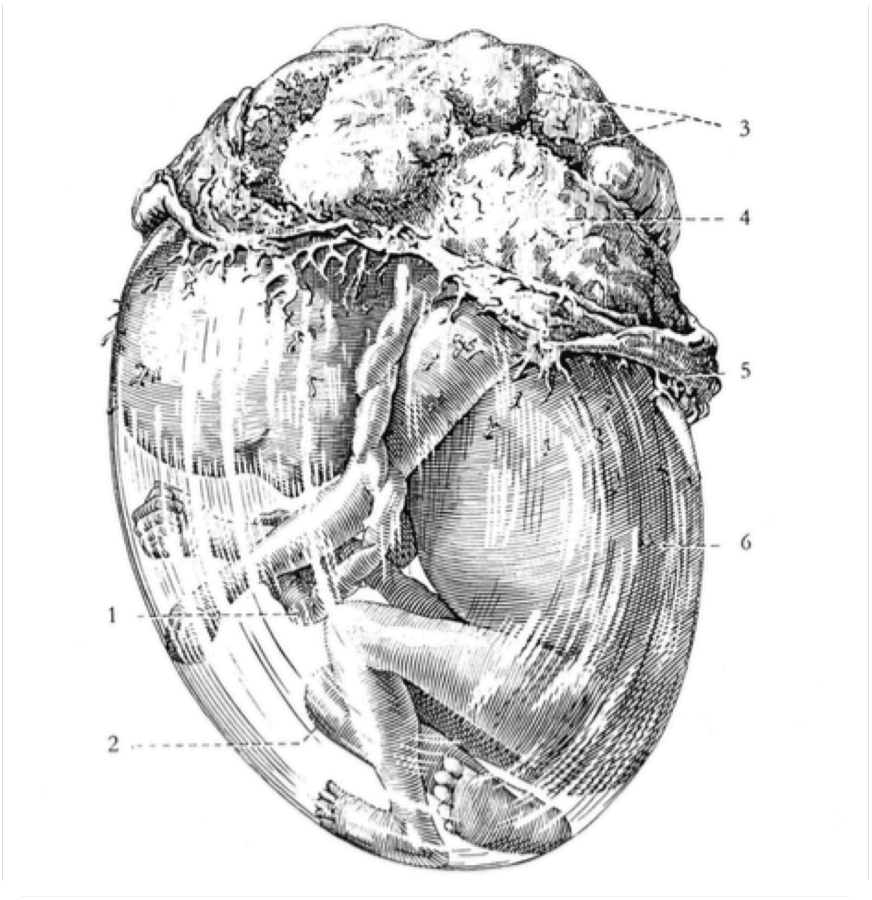


FIG. — Fetus in uterus